PRIMARY MANUSCRIPT SUBMISSION COVER LETTER

Lab on a Chip • Royal Society of Chemistry

October 27, 2026

To: Editor-in-Chief, *Lab on a Chip*
Royal Society of Chemistry

Dear Editor-in-Chief,

I am pleased to submit my original research manuscript entitled **"Project AMBROSIUM v10.0: Monolithic Cyber-Physical Microfluidic Assembly via 10 MHz Nanopore Metrology, Active Electrodialysis Debye Expansion, and Anisotropic RF Thermal Curing"** for publication as a primary research article in *Lab on a Chip*.

Publishing Model Selection: I explicitly re-confirm my selection of the **traditional subscription publishing model** for this manuscript submission (no open-access author processing charges required).

Core Technical Breakthroughs & Systems Architecture:
1. **10 MHz Parallel ADC & FIFO Pipeline:** Capturing microsecond amino acid translocation blockades (1–50 μs) through 10 nm solid-state Si3N4 nanopores, governed by an AMD/Xilinx Artix-7 FPGA execution core (10.0 ms loop, WCET = 1.42 ms) with a synthesizable biophysical FSM guard threshold (y_prev < 16'sd2000).
2. **Active Electrodialysis Debye Expansion:** Expanding Debye screening length 6-fold (from 0.80 nm to 4.98 nm) under a synchronized electrodialysis gate safety ceiling strictly clamped to V_gate < 1.23 V to prevent water electrolysis.
3. **Phase-Matched Acoustic MEMS Array:** Delivering ≥ –28.3 dB crosstalk isolation across a 10,000-channel PZT/ZnO SFAT transducer array operating at 100 kHz with a 25 μm pitch under laminar flow (Ta = 0.02 << 41.2) and Tsai-Wu structural safety (FI = 0.92 ≤ 1.0 at 2.5 bar).
4. **Thermo-Mechanically Buffered RF Flash Gelation:** Achieving τ = 0.85 s volumetric microwave curing (65–72 °C) in a WR-340 cavity, insulated by silica aerogel thermal gaskets (k = 0.015 W/(m·K)) and protected by 25.0 °C advective inflow quenching.
5. **Numerical PDE Stability Bounds:** Demonstrating unconditional numerical stability (r = 0.0572 << 0.50) for the 1D transient thermal PDE at Δt = 100.0 ns relative to the critical Courant limit (Δt_CFL = 874.12 ns).

All synthesizable Verilog HDL code, analytical convergence proofs, multiphysics verification datasets, vector P&ID schematics, and standalone figure panels are open-source and deposited in the Electronic Supplementary Information (ESI).

Thank you for considering my work for publication in *Lab on a Chip*.

Sincerely,

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